1.03 Coordinate Geometry in the x–y Plane				
1.03a	Straight lines	a) Understand and be able to use the equation of a straight line, including the forms $y = mx + c, y - y_1 = m(x - x_1)$ and $ax + by + c = 0$. *Learners should be able to draw a straight line given its equation and to form the equation given a graph of the line, the gradient and one point on the line, or at least two points on the line.* *Learners should be able to use straight lines to find:* *1. the coordinates of the midpoint of a line segment joining two points,* *2. the distance between two points and* *3. the point of intersection of two lines.*		MC1
1.03b		b) Be able to use the gradient conditions for two straight lines to be parallel or perpendicular. *i.e. For parallel lines $m_1 = m_2$ and for perpendicular lines $m_1 m_2 = -1$.*		
1.03c		c) Be able to use straight line models in a variety of contexts. *These problems may be presented within realistic contexts including average rates of change.*		

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
1.03d 1.03e 1.03f	Circles	d) Understand and be able to use the coordinate geometry of a circle including using the equation of a circle in the form $(x-a)^2+(y-b)^2=r^2$. *Learners should be able to draw a circle given its equation or to form the equation given its centre and radius.* e) Be able to complete the square to find the centre and radius of a circle. f) Be able to use the following circle properties in the context of problems in coordinate geometry: 1. the angle in a semicircle is a right angle, 2. the perpendicular from the centre of a circle to a chord bisects the chord, 3. the radius of a circle at a given point on its circumference is perpendicular to the tangent to the circle at that point. *Learners should also be able to investigate whether or not a line and a circle or two circles intersect.*		MC2
1.03g	Parametric equations of curves		g) Understand and be able to use the parametric equations of curves and be able to convert between cartesian and parametric forms. *Learners should understand the meaning of the terms parameter and parametric equation.* *Includes sketching simple parametric curves.* *See also Section 1.07s.*	MC3

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
1.03h	Parametric equations in context		h) Be able to use parametric equations in modelling in a variety of contexts. *The contexts may be within pure mathematics or in realistic contexts, for example those involving related rates of change.*	MC4